

Effects of Beacon Spoofing Attacks on Various LoRaWAN Network Devices

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Abstract— Internet of Things (IoT) devices increasingly support critical infrastructure and are commonly deployed over low-power wide-area networks (LPWANs). Among these, Long Range Wide Area Network (LoRaWAN) is widely adopted. While LoRaWAN v1.1 addresses many early issues, it introduces a Class B beacon dependency that can be exploited via beacon spoofing, degrading downlink availability. Prior work has demonstrated beacon spoofing on single devices, but the resilience of large LoRaWAN networks under such attacks remains underexplored. This paper presents a reproducible large-scale simulator and an empirical study across multiple network scenarios to evaluate the effect of pingSlot diversity on survivability during beacon spoofing. We formulate and test the null hypothesis that network diversity has no effect on attack outcomes. Using Packet Delivery Ratio (PDR) and Beacon Delivery Ratio (BDR) as primary metrics and analysing Attack-Period behaviour, we find that higher pingSlot diversity consistently increases PDR, indicating improved resilience. Accordingly, we reject the null hypothesis and attribute the benefit to the interaction between pingSlot periodicity and pingSlot offset under desynchronisation pressure. Our simulator enables operators to evaluate policies and configurations against beacon spoofing at scale, and our findings highlight the need to harden beacon mechanisms—both within LoRaWAN and in other LPWANs that rely on unauthenticated timing signals.

Keywords—IoT security; IoT penetration testing; beacon spoofing; LoRaWAN; Class B; LPWAN; Packet Delivery Ratio (PDR); Beacon Delivery Ratio (BDR)

I. INTRODUCTION

Cybercriminals continue to develop increasingly sophisticated attack tools despite advances in enterprise security technologies, such as antivirus tools and firewalls, to detect and stop cyberattacks [1, 2]. According to Ben [3], many strategies to protect against cyberattacks assume that the attacker randomly targets networks. If a network has adequate defensive capabilities, the attacker gives up and moves on to a less complicated target. However, research by Trend Micro and others [4] shows this assumption is no longer viable given the rise of advanced, targeted campaigns. At the same time, IoT deployments now span domains from agriculture to smart cities [17]. For example, the City of Gold Coast (Australia) installed 600+ wireless sensors in Springbrook National Park [3].

With more IoT devices across sectors, the attack surface is growing. Home networks reportedly see an average of 10

attacks per day, and Bitdefender blocks about 2.5 million threats daily (~1,736 per minute). The U.S. IoT market is projected to grow 24.7% by 2030—from \$118.24B (2023) to \$553.92B [5]. Once an IoT device is compromised, an attacker can steal data or, even worse, use it to attack others. There are now 10+ billion IoT devices worldwide [2].

Compared to devices such as the personal computer, IoT devices are relatively new technology. Furthermore, IoT devices are fundamentally different. These devices' primary purpose is to collect and exchange data with internet applications [18]. Furthermore, these devices are scattered around an environment, running only on battery power. Therefore, they must be energy efficient and small. Since these devices are small and energy-efficient, their computing resources are limited. Therefore, these devices do not have enough resources to use standard security features such as advanced cryptography [19].

Even though there is a growing effort to find security solutions, the lack of IoT cybersecurity adoption is prominent. According to an HP study, 80% of vulnerable IoT devices have predictable default passwords [6]. This statistic shows that both users and manufacturers underestimate the importance of securing these devices. While most security vulnerabilities could be attributed to either reason discussed above, the security vulnerability studied in this research is a product of both reasons.

Until recently, no other wireless protocol could match the requirements needed for IoT applications [20]. Conventional radio solutions, such as Wide Area Network (WLAN), Bluetooth, and Z-Waves, could be cost-effective but are not adaptable in long-range coverage. Cellular communication solutions such as 2G, 3G, and 4G can provide a broader coverage but are power inefficient [21]. Therefore, Low-power wide-area Networks (LPWANs) technologies came to fruition. These network protocols work across vast distances while handling thousands of devices. Furthermore, they are designed to be energy efficient, a capability needed for battery-powered sensors. There are several Low-power wide-area Networks (LPWANs) on the market, but NB-IoT, Sigfox, and Long Range Wide Area Network (LoRaWAN) remain the leading technologies [22]. In 2021-2022, the New South Wales government has committed \$48 million to farming technologies, taking the first step in creating a LoRaWAN network [7].

Due to its popularity and flexibility, LoRaWAN is a relatively new protocol and subject numerous protocol vulnerabilities [8-10]. For example, one well-known vulnerability allowed malicious actors to obtain a key to the network by replaying an old message [10]. Several attacks exploited these vulnerabilities. The attacks used were Man-in-the-Middle (MITM), bit flipping, Denial Of Service (DoS) and replay attacks. Bit Flipping is an attack in which the attacker changes a ciphertext resulting in a predictable change of the plaintext [23]. Man-in-the-middle (MITM) is when an attacker intercepts communication to eavesdrop between two devices [24].

With the next iteration of LoRaWAN, which is an IoT network protocol, a new class of devices called Class B devices were introduced in 2014 [25]. Along with this new type of device, a new security vulnerability involving a critical signal called beacons appeared. These beacons can be easily spoofed, allowing malicious actors to deteriorate Class B devices' communication capabilities. Beacons are vital signals, sent by the network, to synchronize Class B devices. Since the beacon signal is neither authenticated nor protected, a malicious actor could send a spoofed beacon that could cause a Denial of Service (DoS) into any Class B device that receives it. This would be an example of a Beacon Spoofing attack. While many papers have investigated whether such an attack could work, none have investigated in-depth the possible damage it could cause [10-14].

Previous research studies have demonstrated these Beacon Spoofing attacks on one device [26]. However, there is no extant research on the nature of the resilience of a large LoRaWAN network coming under a Beacon Spoofing attack [27] [28]. The objective of the current research is, therefore, to study the effects of Beacon Spoofing attacks on various LoRaWAN networks/end devices. LoRaWAN networks used in smart cities, have a diverse set of sensors and devices. Therefore, we investigated whether a network's pingSlot diversity affects the potential damage from a Beacon Spoofing attack. To perform this task, several scenarios were performed where a malicious actor broadcasted a spoofed beacon to more than 100 end devices. Each scenario's network differed, revealing how diverse they were.

This article is organized as follows section 2 discusses security aspects of IoT and LoRaWAN. Section 3 describes the methodology and experiment setup. In section 4, the results of the study are provided and discussed. Finally, in section 5, conclusions are drawn, implications are teased out, and recommendations are made.

II. BACKGROUND

LoRaWAN is one of the most studied protocols because it is open-source and ubiquitous. However, most research investigates vulnerabilities around Class A and Class C devices. This is because Class B devices were officially supported in 2017 for LoRaWAN v1.1 [15]. Since Class B devices are relatively new compared to Class A and Class C devices, research related to them is sparse, especially around their beacons. For all end devices to receive the beacon simultaneously, the beacon frames should be neither encrypted nor authenticated [29]. As discussed before, beacons are vital to Class B operations. Therefore, a malicious actor could spoof beacons to inhibit an end device's capability [30].

For these attacks to be successful, the end device should not retrieve the legitimate beacon. An attacker would need to jam the actual beacon signal. Hessel et al. [11] proved this by making the spoofed beacon jam the legitimate one. Beacon frames do not have physical layer CRCs. Therefore, by adding a jamming payload to the end of a malicious beacon, the end device would have trouble locking onto the legitimate beacon. For a spoofed beacon to be malicious, the attacker could change two fields in the beacon packet: GwSpecific and BeaconTime[10, 11].

A. Spoofed GwSpecific

The GwSpecific field contains the GPS coordinates of the broadcasting gateway. Since the beacon packet is not encrypted, a malicious actor could locate and tamper it to perform several attacks. For example, shutting off the gateway would decrease the network's packet delivery rate (PDR), inhibiting its availability [16]. Yang et al. [10] suggested that without even tampering with a gateway, a malicious actor could cause harm by spoofing the GwSpecific field. As Class B devices do not have any power-hungry modules such as GPS receivers, any location-based services could compromise them. An end device could use the GwSpecific field to notify the Network Server about its location. By spoofing a wrong GwSpecific field, the end device could inform the Network Server that it (the end device) is in an entirely different place. To conclude, by spoofing a beacon with a false GwSpecific field, an attacker could endanger an end device's integrity and availability.

B. Spoofed BeaconTime

If a malicious actor spoofs a beacon with a bad beaconTime, numerous entities would be affected, such as the frequency hopping sequence, a device's synchronisation, and the pingSlot offset.

Frequency Hopping Sequence: For regions using fixed frequency plans such as AU915-928, beacons are transmitted in a hopping pattern [15]. As shown in Equation below:

$$BeaconChannel = \left(\frac{beaconTime}{128} \right) \bmod 8 \quad (1)$$

The beaconTime field is used to calculate the next beacon channel [15]. Therefore, if a malicious actor were to spoof a beaconTime that was a whole beacon period ahead (128 seconds), the end device would try to receive the beacon in the wrong channel. Furthermore, if the network operator did not specify the pingSlot Channel, the end device would open its pingSlot in the same manner as the beacons [15]. Thereby, the malicious actor could alter the beaconTime to make the end device open its windows at the right time but in the wrong channels.

End Device's Synchronisation: For an end device to synchronise with the network, the gateways send a beacon periodically [15]. Once the end device receives the beacon, it extracts the beaconTime from the packet. Thus, if the beaconTime is manipulated, end devices would fail to retrieve the next legitimate beacon signal. As shown in Figure 1, if a malicious actor sends a spoofed beacon signal with a delayed beaconTime, it would shift the entire beacon window. Since the Beacon Reserved is shifted, the end device will not retrieve the next legitimate beacon and any Class B packets. However, the end device could still retrieve packets if it goes into the beaconless mode and the legitimate signal is not jammed.

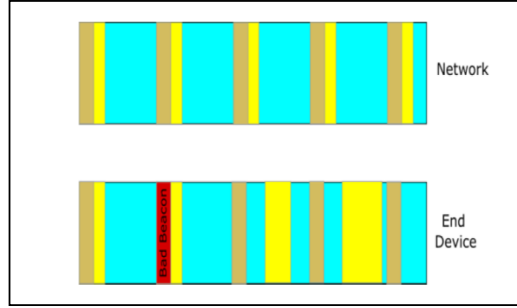


Fig. 1. Beacon Attack on Beacon Reserved

As described before, the Beacon Reserved and pingSlots will expand every beacon period for 120 minutes. If the Beacon Reserved is expanded enough to catch the legitimate signal, the end device will resynchronise with the network. However, if an attacker continuously sends spoofed beacons, the end device will never go into beaconless mode. Therefore, the end device will never synchronise with the network. The attacker could also drain the battery by forcing an end device into the beaconless mode. In that case, the end device could quickly drain its battery if it expands both pingSlots and Beacon Reserved too many times. Yang et al. [10] pointed out that this type of attack on LoRaWAN device would result in labour-intensive expenses to replace the batteries of the end devices.

PingSlot Offset: As discussed before, each end device is given a different pingSlot offset to avoid consistent collisions [15]. As shown in Figure 2, beaconTime is used to calculate the pingSlot offset. Therefore, if an end device receives a malicious beacon with a slightly drifted beaconTime, this could shift the pingSlots by a significant amount. This would allow a malicious actor to disable the downlink communication for one period, without the end device going into the beaconless mode [11]. Since the end device does not go into the beaconless mode, there is no opportunity for the end device to revert to Class A. If the application running on the network does not have a confirmation protocol, this could be a stealthy attack. Es et al. [14] have calculated the effects of a slightly drifted beaconTime on an end device's receiving windows to confirm this attack.

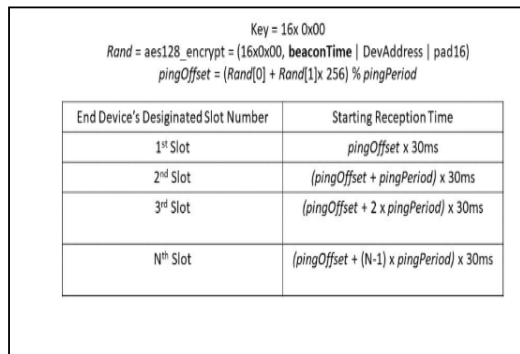


Fig. 2. PingSlot Offset Calculation

However, Es et al. [14] did not study the effects of a spoofed beacon on end devices with different numbers of pingSlots. If there were an increase in pingSlots, what would happen to the downlink communication between the end device and the network? In a real LoRaWAN network, there are multiple end devices with different numbers of pingSlots. If each end device has different numbers of pingSlots, what would happen to the network's overall downlink availability? There is no research studying the effects of spoofed beacons on end devices with different numbers of pingSlots. Therefore, simulation provides a better method through which to study this type of beacon attack.

III. METHODOLOGY

To understand Beacon Spoofing attacks, a research question was developed: What would happen to the network's overall downlink availability if its pingSlot periodicity were changed? To answer the question, two performance evaluation metrics, shown in Table 1, were used. The Beacon Delivery Ratio is used to ensure that both malicious and legitimate beacons are captured by the end device(s). A high Beacon Delivery Ratio indicates that the network operated normally during the Control Period and that the attack worked correctly during the Attack Period. However, a high Beacon Delivery Ratio does not ensure that these two requirements were fulfilled. A high Packet Delivery Ratio in the Control period ensures that the network operates normally, while a low Packet Delivery Ratio during the Attack Period ensures that the attack was successful. Furthermore, the Packet Delivery Ratio is used to analyse an end device/network's behaviour in this experiment.

TABLE I. PERFORMANCE EVALUATION METRIC'S DEFINITIONS

Name	Definition
Beacon Delivery Ratio (%)	The number of beacons received is divided by the total number of beacons sent.
Packets Delivery Ratio (%)	The total number of Class B Packets received is divided by the total number of packets sent.

A. Attack Model

There are several types of Beacon Spoofing attacks used in LoRaWAN networks. For example, Hessel et al. [11] delayed the spoofed beacon slowly to desynchronise an end device. The Beacon Spoofing attacks studied here will directly change the beacon packet (beaconTime). The attacker can spoof a malicious beacon in two ways. The malicious actor could hack a legitimate gateway on the network to spoof malicious beacons (Attack 1). The second method is to use an attacker's own equipment to spoof a beacon (Attack 2). For each attack, the sections below discuss the attacker's objective, what information the attacker needs, what equipment/access control the attacker needs, and how the attack will commence.

B. Experimental Setup

Originally, Attack 2 was going to be simulated because it was the easier of the two for an attacker to perform. However, rather than the gateway making its own beacon, the simulator requires the Network Server to create the beacon packet. Since the gateways do not create the beacon packets, Attack 1 was simulated. Therefore, the simulated Network Server was

modified to add a configurable delay to the beaconTime that all the gateways would use. Once the attack commences (tm), all legitimate gateways start sending malicious beacons.

C. The Assumptions Made by the Study

To analyse complex models, such as LoRaWAN networks/devices, simulation was used. Since simulations cannot replicate real world scenarios exactly, a few assumptions were made. These assumptions are stated below:

Even though beacons are used to correct the end device's internal clocks, there was no simulated time drift for the end devices. This is because time drift could affect an end device's communication. It is noteworthy that the research is only analysing communication loss caused by Beacon Spoofing attacks.

No interference or distance loss model has been introduced into the simulations. Like in the justification of the first assumption, the communication loss caused by interference or distance loss models will affect the research's results.

The end devices are static throughout the simulation to avoid downlink routing conflicts. Identical to the first assumptions justification, the research will only analyse communication caused by Beacon Spoofing attacks.

The study presumes that higher pingSlot periodicity means that an end device requires more downlink packets. Since end devices with higher pingSlot Periodicity (pSp) require more power, sending a small number of packets to a higher pingSlot Periodicity (pSp) end device would be inefficient. Class B packet generation was determined by the end device's pingSlot periodicity (pSp) and a configurable parameter discussed below.

The number of end devices for each gateway should be equal. Since gateways can only send one packet at a time, the number of end devices affects a gateway's throughput.

IV. RESULTS AND DISCUSSIONS

To understand the effects of an attack on a network, we investigated whether a network's end device diversity affects the damage caused by a Beacon Spoofing attack. The following hypothesis was developed: Networks with differing pingSlot periodicity diversity will behave in the same manner during an attack. To test this hypothesis, the study performed eight simulation scenarios. Each scenario had a more diverse group of end devices than the previous scenario. All scenarios had 100 simulation runs each. From each simulation run, the simulator generated information about each packet's sent and received times. Also, the simulator logged the number of beacons sent to and captured by the end device. Finally, each scenario had 100 simulation runs where the Packet Delivery Ratio was averaged together.

In each simulation run, the end device was placed randomly within a disc of radius 5000m. As shown in Figure 3, a few gateways have more end devices with higher pingSlot periodicities. Therefore, a few gateways transmitted more packets than others, leading to traffic congestion. However, this issue was mitigated since the end devices' positions are randomised for each run.

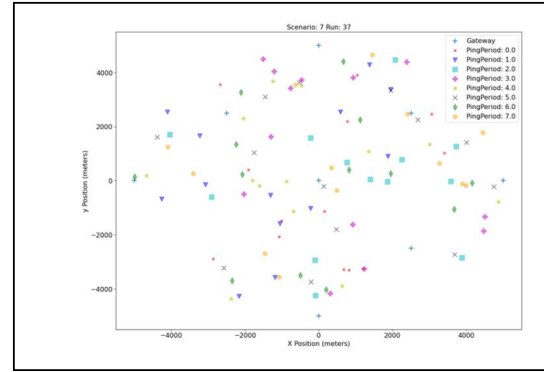


Fig. 3. Example of Scenario 7's Network Topology

In all simulations, all end devices must receive over 90% of packets to be considered operating normally in the Control Period. While in the Attack Period, the Beacon Spoofing attack must deteriorate the network's overall downlink communication by the end of the run. While any decrease in network reliability is bad for a smart city application, the study considered any Packet Delivery Ratio below 80% as detrimental. Therefore, by the end of each run, the Beacon Spoofing attack must deteriorate the network's Packet Delivery Ratio by 20%. Also, all end devices must capture most of the beacons in both the Control and Attack Periods. After evaluating these prerequisites, we performed a regression analysis to evaluate the hypothesis.

A. Prerequisites of Simulation Run

Since beacons are imperative to the operation of the network and the Beacon Spoofing attack, the Average Beacon Delivery Ratio (%) was used to evaluate the first prerequisite. Since downlink communication is a good indication of the last two prerequisites, the Average Packet Delivery Ratio (%) was analysed.

Beacon Transmission Rate: Before evaluating the hypothesis, both the network and attack must operate successfully in each run. An indication of this requirement is the ability of the network to capture beacons. If all end devices in a network are capturing the most legitimate beacons, then the network, as a whole, could operate in Class B properly. While under attack, if most of the end devices are capturing malicious beacons, the attack could succeed. Therefore, if the end devices fail to capture the most legitimate and malicious beacons, it will operate in a beaconless mode, interfering with the results. To sum up, all end devices must capture 95% of the beacons on average in each scenario.

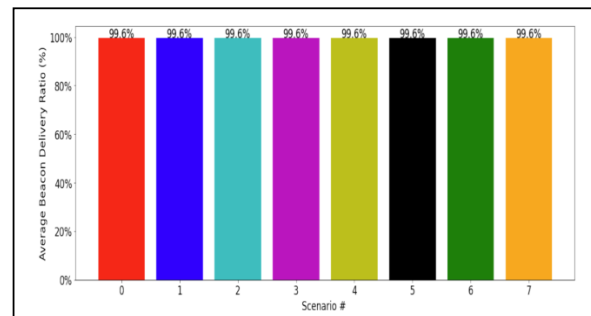


Fig. 4. Average Beacon Delivery Ratio by Scenario

As shown in Figure 4, the end devices received more than 99% of all beacons in all scenarios. Therefore, all scenarios fulfil the first prerequisite. Moreover, the network could operate normally in the Control Period and the cyber-attack could succeed. Though most beacons were captured, this is no guarantee that the network or attack succeeded in each run. Therefore, the Packet Delivery Ratio was analysed.

Overall Network's Downlink Communication: To fulfil the first prerequisite, the network's downlink communication must succeed in the Control Period. Therefore, the network must receive over 90% of all packets. As shown in Figure 5, the average Packet Delivery Ratio in the Control Period is above 90% in all scenarios. Therefore, we can conclude that each scenario's network receives most packets, thereby operating normally in the Control Period. To fulfil the second prerequisite, the Beacon Spoofing attack must severely deteriorate the network's downlink communication. As shown in Figure 5, the cyber-attack decreases the average Packet Delivery Ratio to around 30% for all scenarios. Furthermore, since 1/3 of all packets were sent during the Control Period, a 30% Packet Delivery Ratio means that an insignificant number of packets were received during the Attack Period. Therefore, it is possible to conclude that the Beacon Spoofing attack had substantially damaged the network's downlink availability in all scenarios.

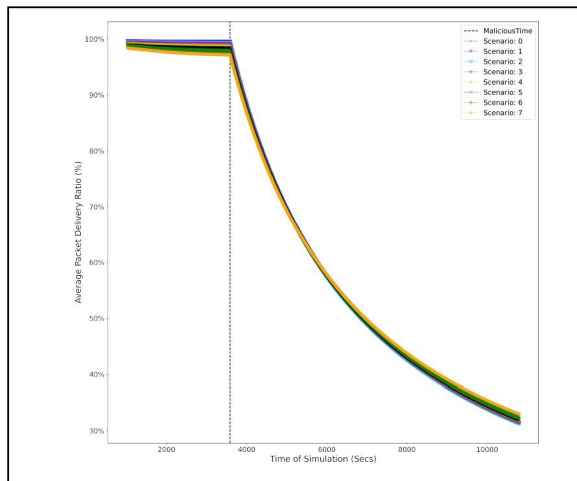


Fig. 5. Average Packet Delivery Ratio by Scenario

B. Evaluating the Hypothesis

Study's hypothesis was that networks with differing pingSlot periodicity diversity will behave in the same manner during an attack. To evaluate this hypothesis, a regression model for the first scenario's Attack Period data was created and compared against other scenarios using two statistical methods. While the regression captured general trends across scenarios, further analysis revealed that Packet Delivery Ratios (PDR) varied with increasing pingSlot diversity. In particular, networks with higher diversity showed slightly improved resilience during beacon spoofing, as they maintained higher PDR values under attack. Therefore, the hypothesis is rejected: network diversity does influence attack outcomes, and greater diversity contributes to improved device survivability.

A regression analysis was used to evaluate the above hypothesis. As shown in Figure 6, all the scenarios had a positively skewed distribution. Therefore, the study created and compared two models: a generalised linear regression model (GLM) and a linear regression model (LM). The independent variable of both models was Simulation Time (Secs), while Average Packet Delivery Ratio (%) was the dependent variable. Both models were created using statsmodel's "Ordinary Least Square" (OLS) and "Generalised Linear Model" (GLM) regression [12]. To determine the best model, two statistical analyses were performed.

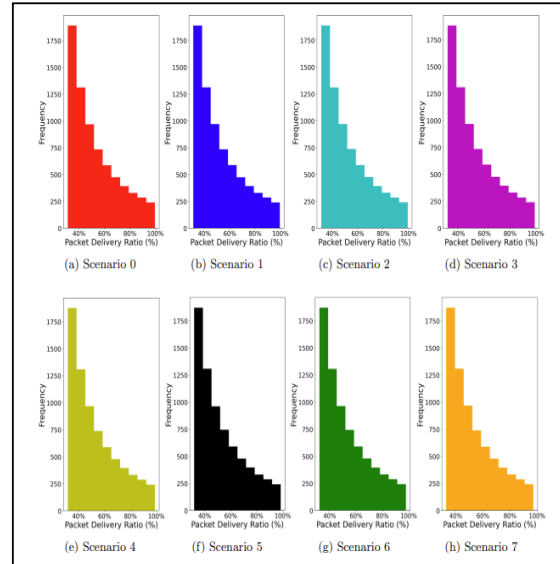


Fig. 6. Top: Distribution Models by Scenarios

Two models were created to determine the best representation of Scenario 0's network behaviour: a generalised linear regression model (GLM) or a linear regression model (LM). As shown in Figure 7, Scenario 0's data have a significantly skewed distribution above zero, mirroring a Gamma distribution. Furthermore, Scenario 0's data mirror an exponential or inverse power relationship, shown in Figure 8. Therefore, using Python's statsmodel library, the GLM was created with gamma as the distribution and inverse power as the link function. The LM model was created with the statsmodel's OLS function.

The best model was chosen based on its coefficient of determination (R^2) and Akaike information criterion (AIC). As shown in Figure 8, due to its higher R^2 and lower AIC, the GLM is surmised to predict Scenario 0's behaviour better than the LM. Therefore, the GLM was used to predict Scenario 0's network behaviour during an attack.

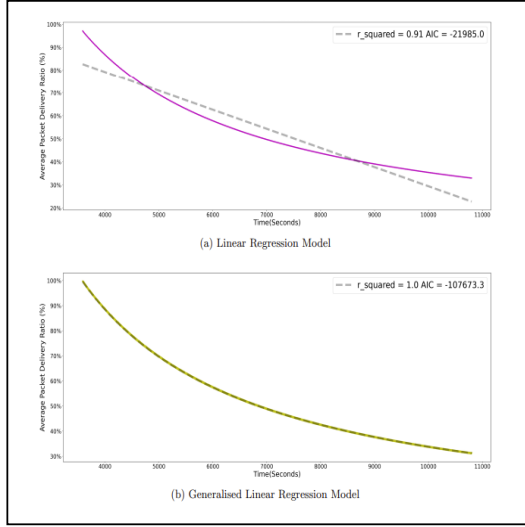


Fig. 7. Distributions (histograms) of PDR across scenarios

Data Comparing Scenario 0's model with other Scenarios' Data: To determine whether pSp diversity could affect a network's behaviour during a Beacon Spoofing attack, Scenario 0's model was cross validated with the data drawn from other scenarios. In Study A, no outside factors affected the end device besides the cyber-attack. However, in Study B, there could be traffic congestion that could slightly affect the results. Therefore, as a preliminary check, we compared each scenario's R^2 to Scenario 0's model (tolerance ≤ 0.10). Although the model captured broad trends across scenarios, decreasing R^2 for higher-diversity settings (Scenarios 6–7), As shown in Figure 8, indicated systematic deviation. This prompted re-calculation of Packet Delivery Ratio for all scenarios restricted to the Attack Period, to test whether pSp diversity influences resilience under beacon spoofing.

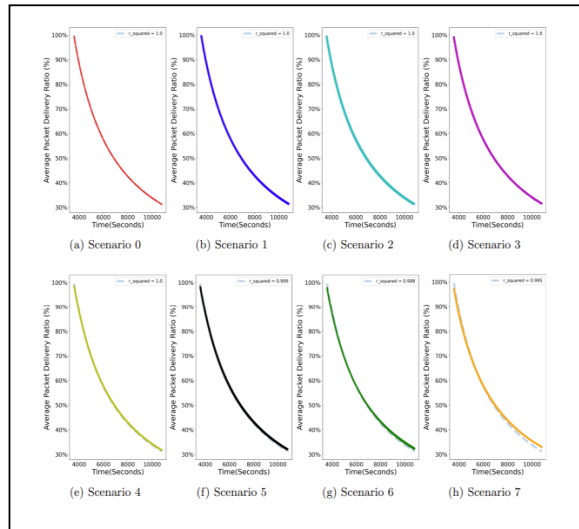


Fig. 8. Top: Distribution Models of All Scenarios

C. Further Investigation

While the preliminary regression suggested broad similarity, the deviations noted in higher-diversity scenarios warranted deeper analysis. We therefore investigated network behaviour during the Attack Period only, isolating the effect of pSp diversity on resilience.

Data Network's Downlink Communication during Attack Period: To investigate the network's behaviour during an attack further, the Packet Delivery Ratio was calculated without the Control Period data. Figure 9 shows the number of packets successfully delivered during the attack. During the Attack Period, Packet Delivery Ratio (PDR) increased monotonically with pingSlot diversity across scenarios. This effect persisted in a per-group analysis of pingSlot periodicity (pSp), confirming that greater diversity systematically improves downlink availability under beacon spoofing.

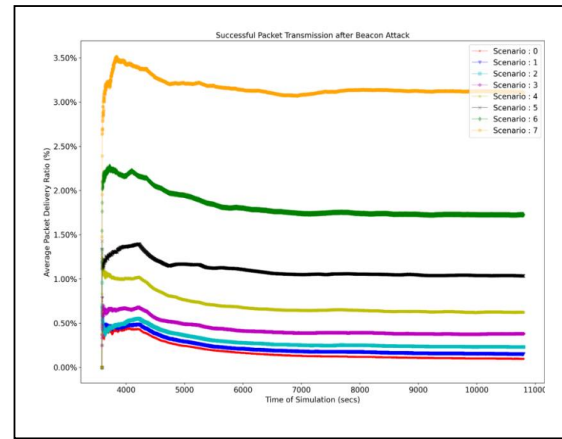


Fig. 9. Packet Delivery Ratio for each Scenario During Attack Period

V. FUTURE WORK

The goal of this study was to determine the effects of a Beacon Spoofing attack on Long Range Wide Area Network (LoRaWAN) networks and end devices. While we have accomplished this basic goal, there is much more to be achieved. All three studies only measured the number of packets and beacons captured. Therefore, adding other performance metrics could expand the investigation into Beacon Spoofing attacks.

- Drain an end device's battery by sending it into a beaconless mode. Therefore, a potential step would be to investigate the amount of power an end device consumes during an attack.
- Another potential project would be to record the number of packet collisions an end device experiences during an attack. This could lead to a possible method of detection of these attacks. In this study, not only were performance metrics limited, but also, the simulations themselves.
- Several simulation scenarios were not explored. For example, future research should study the effects of Beacon Spoofing attacks on channel hopping frequency plans such as AU915. Since the channel

hopping technique uses beaconTime to calculate the subsequent channel frequency, an attack could interfere with this technique.

- d) Another project would be to expand Study C's setup to a larger network. This could enhance our understanding on how the pingSlot offset interferes with Beacon Spoofing attacks.

VI. CONCLUSION

With the emergence of many cutting-edge industries such as smart cities and smart medicine, incorporating Long Range Wide Area Network (LoRaWAN) networks to secure the devices used in these industries has become imperative. Therefore, it is a cause for concern that many researchers often overlook Beacon Spoofing attacks in LoRaWAN Class B devices. By investigating the devastating effects of a Beacon Spoofing attack, this research showcased how important it is to mitigate such attacks. Not only did the study reveal the damaging effects of a Beacon Spoofing attack, but also demonstrated how changing a small parameter could increase an end device's survivability. Consequently, we reject the null hypothesis that networks with differing pingSlot periodicity diversity behave the same under attack. Higher pingSlot diversity improves device survivability and downlink availability during beacon spoofing, and we attribute this effect to the interaction between pingSlot periodicity and pingSlot offset.

As of now, this study implements the first cyber-attack simulator for large LoRaWAN networks. Since experimentation in real-life systems is dangerous and nearly impossible, testing environments with reproducible results has become necessary. Therefore, this simulator could be used to test existing network setups and policies against Beacon Spoofing attacks. It could also be used to test network prototypes. Furthermore, this simulator could be used to train IT personnel in a safe environment where mistakes cause minimal damage. While the primary use of this simulator is for testing cyberattacks, it could also be utilized for network performance testing.

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